

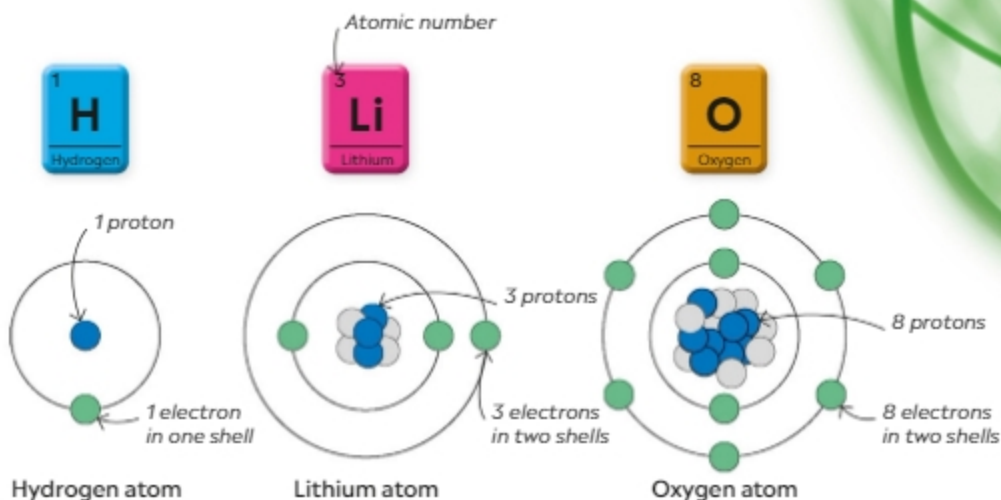
Inside an atom

Everything we see around us is made of very small particles called atoms. Atoms are so tiny that they cannot be observed with the naked eye. They are the smallest units of any element, but the atoms themselves are made up of even smaller “subatomic” particles called protons, electrons, and neutrons. The number of protons in an atom of an element is unique to that element.

The dense nucleus at the center of the atom is where nearly all the mass of an atom lies.

Subatomic particles

Protons and neutrons are found in the core, or nucleus, at the center of the atom, while electrons orbit the nucleus. Protons and electrons have exactly equal but opposite charges; protons are positive, and electrons are negative. Neutrons carry no charge. Because atoms have an equal number of protons and electrons, and neutrons contribute no charge, atoms in their natural state are neutral. Protons and neutrons have the same mass, while electrons are about 10,000 times smaller.



What is the atomic number?

The atomic number of an element tells us how many protons are found inside the nucleus of a single atom of that element. For example, a lithium atom (above) has three protons in its nucleus, which means its atomic number is 3. The elements are arranged on the periodic table in the increasing order of their atomic number, starting with hydrogen, the simplest element, with just one proton.

Electrons travel around the nucleus in three-dimensional areas of space called orbitals.

The orbitals are arranged around the nucleus, at various distances, in layers called shells.